

FPGA Fabric Design

VSD (VLSI System Design)



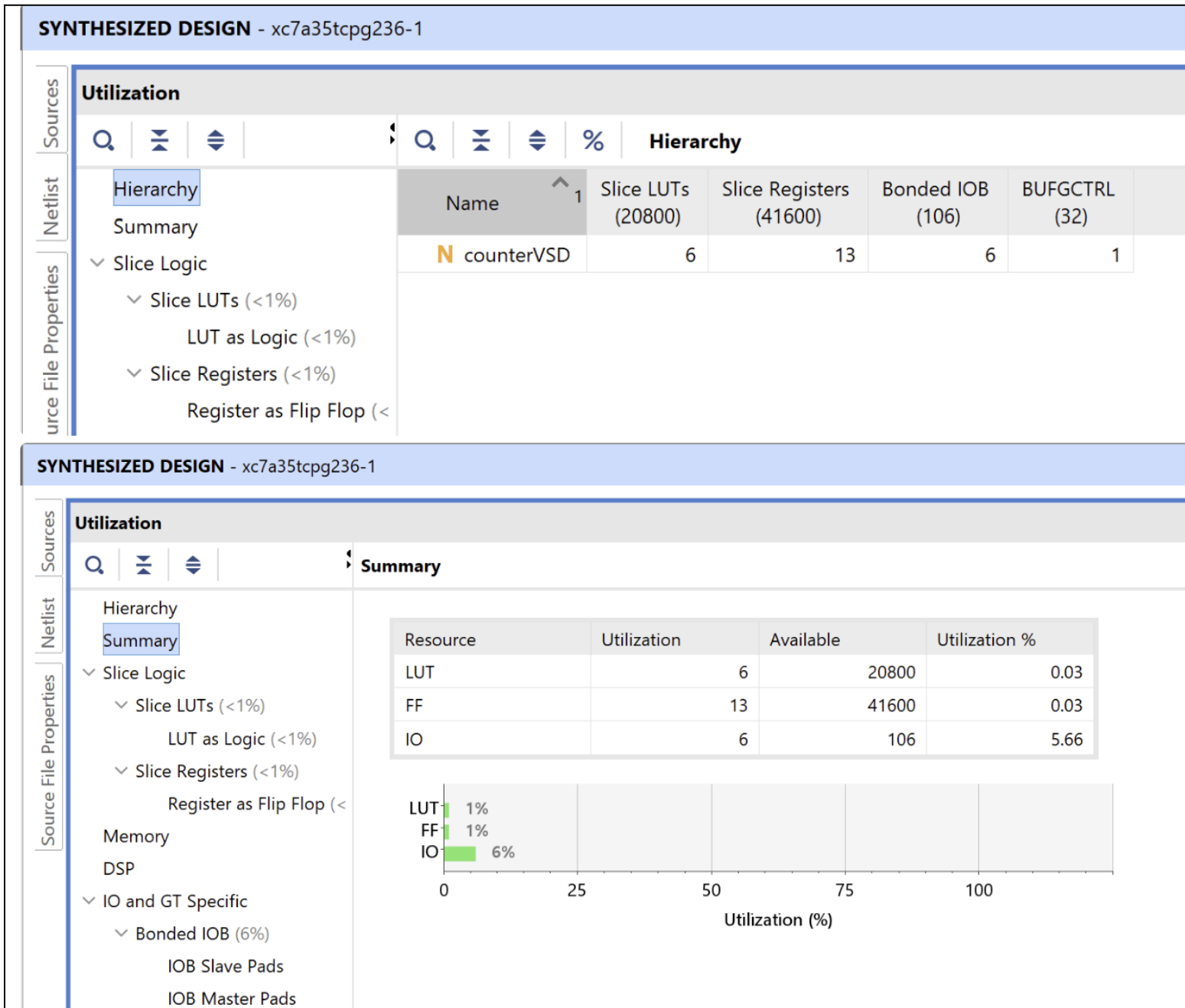
Submitted by Isha Singh

Counter design using Verilog HDL in Vivado

4-bit up-counter

Report

1. Utilization report



2. Power Report

Tcl ConsoleMessagesLogReportsDesign RunsMethodologyDRCPower × Timing

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Summary

Settings

Summary (0.083 W, Margin: N/A)

Power Supply

Utilization Details

- Hierarchical (0.012 W)
- Clocks (<0.001 W)
- Signals (<0.001 W)
 - Data (<0.001 W)
 - Set/Reset (<0.001 W)
- Logic (<0.001 W)
- I/O (0.011 W)

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power:0.083 W

Design Power Budget:Not Specified

Process:typical

Power Budget Margin:N/A

Junction Temperature:25.4°C

Thermal Margin:59.6°C (11.9 W)

Ambient Temperature:25.0 °C

Effective θ JA:5.0°C/W

Power supplied to off-chip devices: 0 W

Confidence level:Low

[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power

14%

86%

Dynamic:0.012 W (14%)

Clocks:<0.001 W (4%)

Signals:<0.001 W (1%)

Logic:<0.001 W (1%)

I/O:0.011 W (94%)

Device Static:0.072 W (86%)

3. Timing report

SYNTHESIZED DESIGN - xc7a35tcbg236-1
?

Sources x Netlist ? _ □ □

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?
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Design Sources (1)

● counterVSD (counterVSD.v)

Constraints (1)

constrs_1 (1)

📄 constraints.xdc (target)

Simulation Sources (1)

sim_1 (1)

Hierarchy

Libraries

Compile Order

Source File Properties ? _ □ □ x

constraints.xdc
⬅ ➡ ⚙

☒ Enabled

Location: C:/Users/shail/Documents/counterVSD/counterVSD.srcs

▼

...

General

Properties

Project Summary x Device x counterVSD.v x counterVSD_tb.v x **constraints.xdc** x ? _ □ □

C:/Users/shail/Documents/counterVSD/counterVSD.srcs/constrs_1/new/constraints.xdc
x

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⬅ ➡
✂
📋
📄
✖
//
🔍
💡
⚙

```

1  set_property IOSTANDARD LVCMOS33 [get_ports {counter_out[2]}]
2  set_property IOSTANDARD LVCMOS33 [get_ports {counter_out[1]}]
3  set_property IOSTANDARD LVCMOS33 [get_ports {counter_out[0]}]
4  set_property IOSTANDARD LVCMOS33 [get_ports {counter_out[3]}]
5  set_property IOSTANDARD LVCMOS33 [get_ports clk]
6  set_property IOSTANDARD LVCMOS33 [get_ports rst]
7  set_property PACKAGE_PIN W5 [get_ports clk]
8  set_property PACKAGE_PIN R2 [get_ports rst]
9  set_property PACKAGE_PIN L1 [get_ports {counter_out[3]}]
10 set_property PACKAGE_PIN P1 [get_ports {counter_out[2]}]
11 set_property PACKAGE_PIN N3 [get_ports {counter_out[1]}]
12 set_property PACKAGE_PIN P3 [get_ports {counter_out[0]}]
13
14 create_clock -period 10.000 -name clk -waveform {0.000 5.000} [get_ports clk]
15

```

Tcl Console
Messages
Log
Reports
Design Runs
Timing
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General Information

Timer Settings

Design Timing Summary

Clock Summary (1)

Methodology Summary

> Check Timing (17)

> Intra-Clock Paths

Inter-Clock Paths

Setup

Hold

Pulse Width

Worst Negative Slack (WNS): 6.116 ns	Worst Hold Slack (WHS): 0.117 ns	Worst Pulse Width Slack (WPWS): 4.500 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 17	Total Number of Endpoints: 17	Total Number of Endpoints: 10

All user specified timing constraints are met.

Timing Summary - timing_1